

Title

Student Academic Performance Analysis Using Data Cleaning and Visualization

1. Introduction

Educational institutions generate large amounts of student-related data that can provide valuable insights into academic performance. However, raw datasets often contain inconsistencies and require preprocessing before meaningful analysis can be performed.

This project focuses on cleaning, processing, and visualizing student academic performance data using Python libraries such as Pandas, Matplotlib, and Seaborn.

The objective is to identify factors influencing student performance and present insights through effective visualizations.

2. Objectives

- Clean and preprocess raw student data
 - Handle missing values and duplicates
 - Analyze academic trends
 - Visualize performance metrics
 - Generate meaningful educational insights
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3. Tools & Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - Matplotlib
 - Seaborn
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4. Dataset Description

The dataset contains:

- Student demographics
- Attendance records
- Study habits
- Previous scores
- Final examination performance

Important Columns:

- Gender
 - Age
 - Study Hours Per Week
 - Attendance Rate
 - Parent Education
 - Internet Access
 - Previous Score
 - Final Score
 - Passed/Failed
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5. Data Cleaning Process

The following preprocessing steps were performed:

- Removed duplicate records
 - Handled missing values
 - Checked data types
 - Corrected inconsistent entries
 - Prepared dataset for visualization
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6. Data Visualization & Analysis

Visualizations used:

- Count plots
- Histograms
- Heatmaps
- Scatter plots

- Bar charts

These visualizations helped identify relationships between study patterns and academic performance.

7. Findings

- Study hours strongly impact academic success.
 - Attendance positively influences marks.
 - Internet access improves learning outcomes.
 - Parent education level affects student performance.
 - Gender performance differences are minimal.
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8. Conclusion

The project successfully demonstrated how data cleaning and visualization techniques can transform raw educational data into actionable insights.

The analysis revealed that study habits, attendance, and access to educational resources significantly affect student performance.

This project highlights the importance of data-driven decision-making in education systems.
